Fiscal Year:	FY 2013	**Task Last Updated:**	FY 01/16/2014
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	RNA Binding Proteins as Evolutionarily Conserved Cellular Spaceflight Response Mechanisms		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	Cheryl.Nickerson@asu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	480-727-7520
Organization Name:	Arizona State University		
PI Address 1:	Center for Infectious Diseases and Vaccinology/The Biodesign Institute		
PI Address 2:	1001 S McAllister Avenue		
PI Web Page:	https://		
City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/01/2009	**End Date:**	04/30/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131 NOTE: extended to 4/30/2013 per NSSC and A.Chu/ARC (Ed., 4/23/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ott, C. Mark (NASA Johnson Space Center) Wilson, James (Villanova University)		
Grant/Contract No.:	NNX09AH40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR'S NOTE: Ground and Flight portions combined into one project per Ames Research Center (July 2010) Understanding infectious disease risks during spaceflight is critical to provide safe passage for human exploration to the moon and Mars. This issue is especially important given reports that the crew's immune system is dysfunctional during flight, and our recent discovery that spaceflight increased the disease-causing potential and globally altered the gene expression of the human food-borne pathogen Salmonella typhimurium. Our research also showed that many of the observed changes in Salmonella were associated with specific proteins in the cells, called RNA binding proteins, that bind to and regulate genetic material within the cell to control cellular characteristics in response to stress. In addition, our studies during spaceflight and using ground-based, spaceflight analogue culture systems have shown that increasing the concentration of certain ionic salts, like phosphates, in the growth media can prevent the increased disease causing potential observed for Salmonella during spaceflight. Interestingly, human cells also contain evolutionarily conserved RNA binding proteins that are very similar in structure and function to those found in bacteria. In human cells, these RNA binding proteins are critical for the cells proper function, including supporting a healthy immune system, maintaining tissue differentiation, and responding to cellular stress. We hypothesize that RNA binding regulatory proteins and their small RNA binding counterparts are key to a conserved, common cellular spaceflight response mechanism in both bacterial and human cells and that this response can be manipulated by environmental salt/ion levels. The implications of a universally conserved molecular response to spaceflight would affect NASA's approach to infectious disease risk assessment, development of biological processing systems for exploration, and changes in biodegradation during a mission. Indeed, if these responses are also observed in human cells, they may translate into knowledge of both the astronaut's response to infection and possibly larger physiological responses that are influenced by cellular changes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will provide novel insight into the mechanisms of the host-pathogen interaction important for disease progression and lay a solid foundation for the development of vaccines and other novel countermeasures, which are not achievable by any other ground-based means, for the treatment and prevention of infectious diseases occurring both on Earth and during spaceflight.
Task Progress:	I) We have identified common evolutionarily conserved spaceflight and spaceflight analogue-induced mechanisms that are conserved across different cell types and which alter their cellular characteristics and phenotypes. Specifically, we have shown that Hfq is an evolutionarily conserved central regulator of the microbial response to both spaceflight and spaceflight analogue culture in both Gram negative (Salmonella typhimurium and Pseudomonas aeruginosa) and Gram positive (Staphylococcus aureus) pathogens. Thus, Hfq represents the first spaceflight and spaceflight-analogue induced regulator acting across bacterial species. In addition, we have also shown that these culture environments regulate the virulence and/or pathogenesis related phenotypes of these same organisms in novel ways that are not observed when they are cultured under conventional conditions. II) A NASA-funded flight experiment on STS-131, operationally called STL-IMMUNE, was awarded as a part of this ground-based flight analogue research. The objectives of this flight experiment were directly related to and complemented our ground-based work (including use of the same cell types), the goal of which is to identify universally conserved molecular responses to ground-based spaceflight analogue culture across a variety of bacterial and mammalian cell types. Specifically, the focus of STL-IMMUNE was to 1) understand the effect of microgravity on the host-pathogen interaction and infection process from the perspective of the human cell, and 2) understand the effect of microgravity alone on human cellular responses that are relevant to normal cellular homeostasis or transition to disease. STL-IMMUNE was the first space biology experiment to conduct an in-flight infection of human cells. We have now finalized sample analysis for global alterations in both transcriptomic and proteomic profiling, cellular differentiation, immune function (inflammatory response mediators), and stress responses associated with spaceflight in the presence and absence of infection. All flight sample comparisons were made to identical synchronous ground controls housed in duplicate CCM hardware as that used for flight. Through iTRAQ analysis, we have identified 86 total unique human proteins (95% confidence, 1% FDR) that were expressed by the HT-29 intestinal epithelial cells cultured on orbit either before or after infection with Salmonella Typhimurium. These proteins belonged to a range of different functional classes and categories, including those important for metabolism, apoptosis, cell communication, immune function, and others. Of the 89 proteins identified, a unique subset of these were found to be differentially regulated in response to spaceflight, and showed differences in response to infection between flight and ground cultures. Collectively, we found intriguing differences in how microgravity uniquely impacted the spaceflight infected intestinal cells as compared to the ground controls. We have also finalized global proteomic response profiling for the A549 lung epithelial cells as well. Moreover, immunohistochemical profiling has revealed differences in both distribution and expression patterns of targeted host cell proteins and alterations in cellular cytoarchitecture. A manuscript reporting these findings will be submitted in early 2014.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	De Weirdt R, Crabbe A, Roos S, Vollenweider S, Lacroix C, van Pijkeren JP, Britton RA, Sarker S, Van de Wiele T, Nickerson CA. "Glycerol supplementation enhances L. reuteri's protective effect against S. Typhimurium colonization in a 3-D model of colonic epithelium." PLoS One. 2012;7(5):e37116. Epub 2012 May 31. http://dx.doi.org/10.1371/journal.pone.0037116 ; PMID: 22693569 , Jun-2012
Articles in Peer-reviewed Journals	Crabbe A, Nielsen-Preiss S, Woolley CM, Barrila J, Buchanan K, McCracken J, Inglis DO, Searles SC, Nelman-Gonzalez MA, Ott CM, Wilson JW, Pierson DL, Stefanyshyn-Piper HM, Human LE, Nickerson CA. "Spaceflight enhances cell aggregation and random budding in Candida albicans." PLOS One. 2013;8(12): e80677. Published 04 Dec 2013. http://dx.doi.org/10.1371/journal.pone.0080677 ; PMID: 24324620 , Dec-2013
Articles in Peer-reviewed Journals	Kennedy AR, Crucian B, Huff J, Klein S, Moren D, Murasko D, Nickerson CA, Sonnenfeld G. "Sex-Based Differences in the Effects of Space Flight on the Immune System." Journal of Women's Health. 2013; Accepted. , Dec-2013

Articles in Peer-reviewed Journals	Soni A, O'Sullivan L, Quick LN, Ott CM, Nickerson CA, Wilson JW. "Conservation of the low-shear modeled microgravity response in Enterobacteriaceae and analysis of the trp genes in this response." Open Microbiol J. 2014 Jun 13;8:51-8. eCollection 2014. http://dx.doi.org/10.2174/1874285801408010051 ; PubMed PMID: 25006354; PubMed Central PMCID: PMC4085587 , Jun-2014
Articles in Peer-reviewed Journals	Yang J, Barrila J, Roland KL, Kilbourne J, Ott CM, Forsyth RJ, Nickerson CA. "Characterization of the invasive, multidrug resistant non-typhoidal Salmonella strain D23580 in a murine model of infection." PLoS Negl Trop Dis. 2015 Jun 19;9(6):e0003839. eCollection 2015 Jun. http://dx.doi.org/10.1371/journal.pntd.0003839 ; PubMed PMID: 26091096; PubMed Central PMCID: PMC4474555 , Jun-2015
Articles in Peer-reviewed Journals	Barrila J, Sarker SF, Hansmeier N, Yang S, Buss K, Briones N, Park J, Davis RR, Forsyth RJ, Ott CM, Sato K, Kosnik C, Yang A, Shimoda C, Rayl N, Ly D, Landenberger A, Wilson SD, Yamazaki N, Steel J, Montano C, Halden RU, Cannon T, Castro-Wallace SL, Nickerson CA. "Evaluating the effect of spaceflight on the host-pathogen interaction between human intestinal epithelial cells and Salmonella Typhimurium." npg Microgravity. 2021 Mar 9;7(1):9. https://doi.org/10.1038/s41526-021-00136-w ; PMID: 33750813; PMCID: PMC7943786 , Mar-2021
Books/Book Chapters	Ott CM, Crabbe A, Wilson JW, Barrila J, Nickerson.CA. "Effect of spaceflight on microbial gene expression and virulence." in "Stress Challenges and Immunity in Space: From Mechanisms to Monitoring and Preventive Strategies." Ed. A. Chouker. New York : Springer-Verlag, 2012. p. 203-226. http://www.springer.com/978-3-642-22271-9 , Dec-2012
Significant Media Coverage	ASU BioDesign Institute. "News article about PI's STS-131 experiment, 'Out of this world: new study investigates infection of human cells in space.' " ASU BioDesign Institute News, March 30, 2010. http://www.biodesign.asu.edu/news/out-of-this-world-new-study-investigates-infection-of-human-cells-in-space , Mar-2010
Significant Media Coverage	Space Biosciences Division. NASA Ames Research Center. "News article about collaborative Space Tissue Loss experiment, including Dr. Nickerson's part in STS-131 experiment, 'Space Tissue Loss (STS-131) .' " Online article, April 2010. http://www.nasa.gov/centers/ames/news/releases/2010/10-23AR.html ; accessed 11/24/14., Apr-2010
Significant Media Coverage	O'Brien M. "Spaceflight research highlighted on the PBS News Hour television documentary series with science correspondent Miles O'Brien. Documentary to air in early 2014." PBS News Hour television documentary series with science correspondent Miles O'Brien. Documentary to air in early 2014., Jan-2014
Significant Media Coverage	AAAS Podcast. "Podcast: Science From the International Space Station. AAAS Podcast interview covering Dr. Nickerson's invited presentation in the session Science from the International Space Station, AAAS meetings in Boston, MA, Feb, 2013." AAAS Podcast. February 19, 2013. http://news.sciencemag.org/2013/02/podcast-science-international-space-station?ref=hp , Feb-2013
Significant Media Coverage	ASU BioDesign Institute. "News article about the fourth publication resulting from PI's STS-115 experiment, 'Fungal pathogen shows profound effects from spaceflight', ASU Biodesign Institute, Dec 16, 2013. This is the third pathogen reported from this work and the first report of the effect of spaceflight culture on the global gene expression and phenotypic characteristics of a eukaryotic pathogen." ASU Biodesign Institute, Dec 16, 2013. http://www.biodesign.asu.edu/news/fungal-pathogen-shows-profound-effects-from-spaceflight , Dec-2013